

The Grokking Phenomenon in Neural Networks: A Comprehensive Review of Contemporary Research 2022-2025

arXiv Literature Review by Nikolay Yudin

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Abstract

Grokking is a phenomenon of delayed generalization in neural networks, where a model demonstrates a sharp improvement in test accuracy significantly after achieving near-zero training error. First described in 2022, this effect has attracted considerable attention from the research community and spawned over 80 publications on arXiv. This review systematizes the current understanding of grokking, covering theoretical foundations, mechanistic interpretability, acceleration methods, practical applications, and connections to other fundamental phenomena in deep learning. We trace the evolution of research from the initial discovery on algorithmic tasks to applications in large language models, reinforcement learning, and real-world reasoning tasks. Special attention is given to the mechanisms underlying grokking, including phase transitions, implicit regularization, internal circuit formation, and weight norm minimization on the zero-loss manifold.

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1 Introduction

1.1 Motivation

Traditional understanding of neural network training assumes that generalization occurs in parallel with decreasing training error. However, in 2022, Power et al. [1] discovered a surprising phenomenon: when training transformers on modular arithmetic tasks, models achieved nearly perfect training accuracy while remaining at random guessing levels on the test set for tens of thousands of optimization steps. Then, suddenly and dramatically, test accuracy increased from $\sim 50\%$ to $\sim 100\%$ over a relatively short period.

This phenomenon was named *grokking* — a term borrowed from Robert Heinlein’s science fiction novel “Stranger in a Strange Land”, where it means deep, intuitive understanding. In the context of machine learning, grokking represents delayed generalization occurring far beyond the point of overfitting.

1.2 Definition and Formalization

Definition 1 (Grokking). *Grokking is a phenomenon in neural network training characterized by:*

1. *Rapid achievement of near-zero training error ($\mathcal{L}_{\text{train}} \approx 0$)*
2. *Extended plateau phase where test accuracy remains low ($\text{Acc}_{\text{test}} \approx \text{random}$)*
3. *Sharp phase transition to high test accuracy ($\text{Acc}_{\text{test}} \rightarrow 1$) occurring after prolonged training*

Formally, let $\mathcal{L}_{\text{train}}(t)$ and $\mathcal{L}_{\text{test}}(t)$ denote training and test errors at time t . Grokking is observed when there exist times $t_1 < t_2 < t_3$ such that:

$$\mathcal{L}_{\text{train}}(t_1) \approx 0, \tag{1}$$

$$\mathcal{L}_{\text{test}}(t_2) \approx \mathcal{L}_{\text{random}}, \tag{2}$$

$$\mathcal{L}_{\text{test}}(t_3) \approx 0, \tag{3}$$

where $t_3 - t_1 \gg 0$ (the delay can span thousands or tens of thousands of iterations).

1.3 Historical Perspective

2022: Discovery. Power et al. [1] first described the phenomenon on modular arithmetic tasks. Liu et al. [2] expanded understanding, showing grokking on MNIST and identifying four learning phases.

2023: Theoretical Foundations. Nanda et al. [3] applied mechanistic interpretability, reverse-engineering the algorithm learned by transformers. Davies et al. [4] unified grokking with double descent.

2024: Acceleration and Applications. Methods developed to accelerate grokking by 50+ times [7]. Phenomenon discovered in deep networks [8].

2025: Real-World Applications. Grokking in large language model pretraining [9], reinforcement learning [10], and knowledge graph reasoning tasks [11].

1.4 Review Structure

Section 2 describes theoretical foundations of grokking. Section 3 focuses on mechanistic interpretability. Section 4 examines acceleration methods. Section 5 covers practical applications. Section 6 explores connections to other phenomena. Section 7 concludes the review and outlines future research directions.

2 Theoretical Foundations

2.1 Phase Transitions and Critical Phenomena

Several works interpret grokking as a phase transition in parameter space or training dynamics.

2.1.1 Grokking as First-Order Phase Transition

Work [12] shows that in two-layer networks, grokking can be formalized as a first-order phase transition. Authors propose a mapping between training dynamics and phase transition theory, predicting:

- Critical data size below which grokking does not occur
- Exponential slowdown of dynamics near the critical point
- Bimodal distribution across ensemble of initializations

However, later work [13] disputes this, arguing that grokking is *not* a first-order phase transition but rather computational glass relaxation.

2.1.2 Information-Theoretic Perspective

Clauw et al. [14] use synergy and redundancy measures to characterize grokking as an emergent phase transition. They show that:

$$I(X; Y) = R(X; Y) + S(X; Y), \quad (4)$$

where R is redundancy, S is synergy. During grokking, synergy sharply increases, indicating formation of integrated representations.

2.2 Implicit Regularization and Implicit Bias

2.2.1 Dichotomy of Early and Late Phases

Lyu et al. [15] prove that grokking can be induced by dichotomy of implicit biases in early and late training phases:

Theorem 1 (Informal). *When training homogeneous neural networks with large initialization and small weight decay:*

1. *Early phase: gradient descent gets trapped in solution corresponding to kernel predictor (overfitting)*

2. *Late phase: sharp transition occurs to min-norm/max-margin predictor (generalization)*

Authors show that optimizers may implicitly induce different biases in different phases, and grokking occurs when early phase bias leads to overfitting and late phase bias to generalization.

2.2.2 Role of Weight Decay

Weight decay plays a critical role in grokking:

- **Nanda et al.** [3]: weight decay removes memorization components in cleanup phase
- **Gromov** [5]: grokking can occur without explicit regularization under certain conditions
- **Liu et al.** [6]: mismatch between train and test losses (“LU mechanism”) depends on weight norm

2.3 Loss Landscape Geometry

2.3.1 Norm Minimization on Zero-Loss Manifold

Recent work [16] proposes an elegant explanation:

Theorem 2 (Informal). *After achieving memorization (zero training error), training dynamics approximately follow weight norm minimization on the zero-loss manifold:*

$$\min_{\theta} \|\theta\|^2 \quad s.t. \quad \mathcal{L}_{train}(\theta) = 0 \quad (5)$$

For two-layer networks, authors derive closed-form expressions for the cost function of the embedding layer. This explains why:

1. Grokking requires prolonged training (slow norm minimization)
2. Weight decay accelerates grokking (explicitly penalizes norm)
3. Low-norm solutions generalize better

2.3.2 Riemannian Norm Minimization

Alternative theoretical framework [17] proposes a two-phase picture:

1. **Interpolation:** model rapidly converges to poor global minimum (memorization)
2. **Riemannian norm minimization:** weight decay gradually steers iterates toward lower-norm solution with better generalization

2.4 Transition from Lazy to Rich Learning

2.4.1 Lazy Regime vs. Rich Regime

Work [18] ICLR 2024 proposes that grokking arises from neural networks transitioning from *lazy* training dynamics to *rich* feature learning regime:

Lazy regime:

- Parameters remain close to initialization
- Learning occurs in neural tangent kernel (NTK) regime
- Memorization without structured feature learning

Rich regime:

- Parameters move significantly from initialization
- Active feature learning
- Formation of structured representations

Authors provide evidence for this transition on MNIST, single-layer transformers, and teacher-student networks.

2.4.2 Theoretical Analysis of Modular Addition

Mohamadi et al. [19] provide theoretical explanation for grokking in modular addition:

Theorem 3 (Simplified). *Early in gradient descent when kernel regime holds, no permutation-equivariant model can achieve small population error unless it sees at least a constant fraction of all data points.*

This strongly supports interpretation of grokking as consequence of transition from kernel-like behavior to limiting behavior of gradient descent on deep networks.

2.5 Scaling Laws and Critical Data Size

2.5.1 Provable Scaling Laws

Work [20] proposes $\mathcal{L}_i^2$ framework capturing three key stages of grokking in two-layer nonlinear networks:

1. **Lazy learning:** parameters evolve slowly
2. **Independent feature learning:** each feature learns separately
3. **Interactive feature learning:** features interact

Authors derive provable scaling laws for feature emergence, memorization, and generalization, shedding light on roles of hyperparameters:

- Weight decay γ
- Learning rate η
- Sample size n

2.5.2 Critical Data Size

Liu et al. [21] introduce concept of critical data size in language models — a sharp threshold marking phase transition between memorization and generalization:

$$n_{\text{critical}} \propto \frac{C}{\gamma \cdot \eta}, \quad (6)$$

where C depends on task complexity.

Below n_{critical} , model cannot grok; above this threshold, grokking becomes possible, but time to grok depends on distance to critical point.

2.6 Grokking and Benign Overfitting

Benign overfitting is a phenomenon where a model perfectly fits noisy data but still generalizes optimally. Work [22] shows the connection:

Theorem 4 (Informal). *ReLU networks trained on XOR cluster data with flipped labels can exhibit:*

1. *Benign overfitting: perfect fit + optimal generalization*
2. *Harmful overfitting followed by grokking: overfitting $\rightarrow$ transition $\rightarrow$ optimal generalization*

This is the first theoretical result for benign overfitting in neural network classification when data distribution is not linearly separable.

3 Mechanistic Interpretability

3.1 Reverse-Engineering Algorithms

3.1.1 Progress Measures via Mechanistic Interpretability

Nanda et al. [3] fully reverse-engineered the algorithm learned by transformers for modular addition:

Algorithm:

1. Inputs are encoded via discrete Fourier transform (DFT)
2. Addition is transformed to rotation on a circle
3. Trigonometric identities are used to compute result

Mathematically, for $a + b \bmod p$:

$$\text{DFT}(a + b) = \text{DFT}(a) \cdot \text{DFT}(b), \quad (7)$$

which allows reducing addition to elementary operations in frequency domain.

3.1.2 Three Training Phases

Using mechanistic interpretability, authors identify three continuous phases:

1. Memorization:

- Training error rapidly decreases
- Model memorizes examples without structure
- Test error remains high

2. Circuit Formation:

- Gradual amplification of structured mechanisms
- DFT algorithm forms in weights
- Test error begins slowly decreasing

3. Cleanup:

- Weight decay removes memorization components
- Sharp transition to perfect test accuracy
- Structured mechanisms dominate

3.2 Circuit Efficiency

3.2.1 Generalization vs. Memorization

Varma et al. [23] propose that grokking occurs when tasks admit both generalizing and memorizing solutions, where generalizing solution is slower to learn but more efficient.

Efficiency definition:

$$\text{Efficiency} = \frac{\text{Performance}}{\text{Complexity}}, \quad (8)$$

where complexity can be measured by weight norm, number of parameters, or computational cost.

3.2.2 Novel Phenomena

The work demonstrates:

- **Ungrokking:** degradation after grokking under certain conditions
- **Semi-grokking:** partial generalization improvement
- Application to symbolic manipulation in language models

3.3 Automated Circuit Discovery

Conmy et al. [24] developed ACDC (Automated Circuit DisCovery) algorithm for automated circuit discovery in transformers:

Results:

- Rediscovered all 5 component types in GPT-2 Small circuit for Greater-Than operation
- Selected 68 out of 32,000 edges, all manually found by previous work
- Confirmed that generalizing circuits form gradually during grokking

3.4 Embedding Layer Dynamics

3.4.1 Mechanistic Insights from Embeddings

Work [25] focuses on embedding layer dynamics:

Key findings:

1. **IPR compression:** Inverse Participation Ratio of embeddings drops by $\sim 37\%$ during grokking
2. **Rank compression:** effective dimensionality of embeddings sharply decreases
3. **Structured patterns:** embeddings organize into interpretable structures (e.g., circular patterns for modular arithmetic)

3.5 Neural Collapse

3.5.1 Debates on Neural Collapse

Two works offer opposing views:

Position 1: Neural collapse explains grokking [26]

- Within-class variance contraction is key factor
- Unified explanation of grokking and information bottleneck
- Neural collapse emergence correlates with grokking

Position 2: Flatness is necessary, neural collapse is not [27]

- Only relative flatness consistently predicts generalization
- Models forced to collapse or prevented from collapsing generalize equally well
- Models regularized away from flat solutions exhibit delayed generalization (grokking)

The question remains open and actively debated.

3.6 Grokking vs. Standard Learning

3.6.1 Same Features, Different Encodings

Recent work [28] investigates fundamental differences:

Results:

- Grokked and standard-trained models learn *same features*
- Large differences in *encoding efficiency* of these features
- Peak compressibility achieved immediately after grokking plateau
- Feature development during plateau is task-dependent

Information-geometric measures show that models undergoing grokking follow a straight path in information space.

4 Grokking Acceleration Methods

4.1 Amplifying Slow Gradients

4.1.1 Grokfast

Work [7] proposes elegant method to accelerate grokking by over 50 times:

Core idea: Spectrally decompose parameter trajectories under gradient descent into two components:

1. **Fast-varying component:** leads to overfitting
2. **Slow-varying component:** induces generalization

Algorithm:

$$\theta_{t+1} = \theta_t - \eta \cdot \alpha(g_t), \tag{9}$$

where $\alpha(g_t)$ is operator amplifying slow components of gradient g_t .

Practical details:

- Requires only a few lines of code
- Applicable to diverse tasks: images, language, graphs
- No architecture or loss function changes needed

4.2 Embedding Transfer

4.2.1 Let Me Grok for You

Method [29] accelerates grokking through embedding transfer from weaker model:

Procedure:

1. Train small model until grokking
2. Extract its embeddings

3. Initialize large model with transferred embeddings
4. Train large model (grokking occurs faster)

Theoretical justification:

- Weak model already learned structured features
- Strong model starts with better initialization
- Long circuit formation phase is skipped

4.3 Jacobian Alignment

4.3.1 GrokAlign

Work [30] proves theoretically and demonstrates empirically:

Theorem 5 (Informal). *Aligning network’s Jacobian with training data (in cosine similarity sense) ensures grokking under low-rank Jacobian assumption.*

GrokAlign method:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \cdot \mathcal{L}_{\text{Jacobian}}, \quad (10)$$

where

$$\mathcal{L}_{\text{Jacobian}} = -\frac{1}{n} \sum_{i=1}^n \cos(J_{\theta}(x_i), x_i), \quad (11)$$

$J_{\theta}(x_i)$ is Jacobian of output with respect to input.

Advantages:

- Induces grokking much sooner than weight decay
- Theoretical guarantees under certain conditions
- Applicable to deep networks

4.4 Kolmogorov-Arnold Representation

Work [31] proposes using Kolmogorov-Arnold representation to accelerate grokking in arithmetic operations:

Key methods:

1. **Commutative augmentation:** for commutative operations ($a+b = b+a$) doubles effective data size
2. **Transfer learning:** observing shared structure between KA representations of binary operations

Empirical results:

- Significant acceleration on modular addition/multiplication tasks
- Better interpretability of learned representations

4.5 Data Selection and Augmentation

4.5.1 Grokking Beyond Euclidean Norm

Work [32] shows:

Theorem 6 (Informal). *When there exists a model with property P (e.g., sparse or low-rank weights) that generalizes, gradient descent with small non-zero regularization of P results in grokking.*

Practical corollary: Grokking can be amplified *solely through data selection*, without changing regularization or architecture.

4.5.2 Data Augmentation for Real-World Reasoning

Work [11] extends grokking to real-world factual data:

Approach:

- Augment knowledge graphs with synthetic data
- Raise ratio of inferred facts to atomic facts above grokking threshold
- Apply curriculum learning to accelerate memorization phase

Results:

- 95 – 100% accuracy on 2WikiMultiHopQA
- First demonstration of grokking on real-world multi-hop reasoning tasks

4.6 Knowledge Distillation

Work [33] shows that knowledge distillation from grokked model can:

- Induce grokking on different distribution
- Accelerate grokking even when data is below critical threshold
- Transfer structured representations between models

5 Practical Applications

5.1 Large Language Models

5.1.1 Grokking in LLM Pretraining

Work [9] presents first study of grokking in practical LLM pretraining:

Experimental setup:

- One-epoch pretraining of next-token prediction
- Cross-domain large-scale corpus
- Generalization on diverse benchmark tasks

Key findings:

1. Grokking still occurs in one-pass pretraining of practical-scale LLMs
2. Grokking is *local and asynchronous* across different data groups/domains
3. Pathways evolve from random, non-smooth, instance-specific to more structured and transferable

Implications:

- Possibility of monitoring memorization → generalization transition without test set
- Potential for domain-adaptive training strategies

5.2 Reinforcement Learning

5.2.1 RL Grokking Recipe

Work [10] is first to show grokking during RL fine-tuning of LLMs:

Observations:

- “Hard frontier” exhibits distinct learning dynamics with grokking-like phase transition
- May require hundreds or thousands of training steps per task
- RL unlocks and transfers new algorithms in LLMs through grokking

Significance:

- Explains why RL fine-tuning is effective for complex tasks
- Proposes new metrics for evaluating RL progress

5.3 Knowledge Graphs and Multi-Hop Reasoning

As noted in Section 4.5.2, work [11] successfully applied grokking to:

- Real-world knowledge graphs (Freebase, Wikidata)
- Multi-hop questions requiring compositional reasoning
- Achieving state-of-the-art results on 2WikiMultiHopQA

5.4 Machine Unlearning

5.4.1 Grokked Models are Better Unlearners

Liang et al. [34] discover surprising connection:

Results:

- Grokked models achieve 6 – 8% better unlearning effectiveness
- Maintain 10 – 20% higher performance on retained data

- Grokking fundamentally restructures internal representations for selective forgetting

Mechanism:

1. Grokking creates more structured, decomposed representations
2. These representations allow removing specific information without collateral damage
3. Minimal “collateral damage” to retained data

AI safety implications:

- More practical privacy-preserving ML systems
- Compliance with regulations (GDPR “right to be forgotten”)
- Potential for alignment and debiasing

5.5 Computer Vision

Though grokking research in CV is less extensive, several works showed:

- Grokking on MNIST [2]
- Application in Vision Transformers [40]
- Connection to data augmentation and curriculum learning [35]

6 Connections to Other Phenomena

6.1 Double Descent

6.1.1 Unifying Grokking and Double Descent

Davies et al. [4] propose unified framework:

Hypothesis: Grokking and double descent are instances of same learning dynamics within framework of pattern learning speeds.

Framework:

- Model learns multiple “patterns” at different speeds
- **Sample-wise grokking:** different data subsets grok at different times
- **Model-wise grokking:** different models (by size) grok at different times

First demonstration of model-wise grokking:

- Small models: rapid overfitting, long absence of generalization, then grokking
- Large models: may generalize earlier or grok later

6.2 Benign Overfitting

As discussed in Section 2.6, work [22] shows grokking and benign overfitting are related:

- Both arise when overfitting to noisy data
- Benign overfitting: immediate generalization despite overfitting
- Grokking: delayed generalization after overfitting
- Choice between them depends on initialization, learning rate, model size

6.3 Lottery Ticket Hypothesis

6.3.1 Grokked Tickets

Work [39] connects grokking with lottery ticket hypothesis:

Grokked Tickets concept:

- Lottery tickets obtained during generalization phase (after grokking)
- Significantly reduce delayed generalization on different tasks

Key finding:

$$\text{Good structure} > \text{Weight norm} + \text{Sparsity} \quad (12)$$

Delayed generalization mitigation occurs not only due to reduced weight norm or increased sparsity, but due to *discovery of good subnetworks*.

Experiment:

- Pruning can transfer memorizing network to generalizing one *without weight updates*
- Supports claim that delayed generalization occurs due to discovery of good structure

6.4 Neural Collapse and Flatness

Debates on role of neural collapse vs. flatness were discussed in Section 3.4. Consensus:

- **Flatness:** consistently predicts generalization and grokking
- **Neural collapse:** correlates but is neither necessary nor sufficient
- Regularization away from flatness $\Rightarrow$ delayed generalization (grokking)

6.5 Emergent Abilities

Work [36] proposes unified view of grokking, double descent, and emergent abilities through lens of circuit competition:

Core idea:

- Model contains multiple “circuits” (solution mechanisms)
- Circuits compete during training
- Dominance transition from one circuit to another $\Rightarrow$ performance phase transition

Connection to grokking:

- Memorizing circuit dominates early (learns fast)
- Generalizing circuit slowly forms
- Cleanup phase: generalizing circuit displaces memorizing one

6.6 Edge of Numerical Stability

Work [37] proposes new perspective:

Thesis: Without regularization, grokking tasks push models to edge of numerical stability, introducing floating-point errors in Softmax function.

Solutions:

1. **StableMax:** modified activation function
2. **$\perp$ Grad:** training algorithm for regularization-free grokking

Significance:

- Explains why regularization (weight decay) is so effective
- Proposes alternative methods for inducing grokking
- Connects grokking to numerical properties of optimization

7 Open Questions and Future Directions

7.1 Theoretical Challenges

7.1.1 Necessary and Sufficient Conditions

Work [38] establishes necessary and sufficient conditions for grokking, proving that *uniformity* is key factor in test accuracy improvement. However:

- Complete characterization for general architectures is lacking
- Role of nonlinearities requires further analysis
- Extension to continuous feature spaces needed

7.1.2 Scaling to Modern Architectures

Most theoretical results are for:

- Two-layer networks
- Small tasks (modular arithmetic, MNIST)
- Simplified conditions

Open questions:

1. Does grokking occur in modern LLMs (GPT-4, Claude, etc.) during pretraining?
2. How does grokking scale with model size ($10^9 - 10^{12}$ parameters)?
3. Do theoretical insights apply to Vision Transformers, diffusion models?

7.2 Practical Directions

7.2.1 Hyperparameter Optimization

While several works study hyperparameter influence, systematic understanding is lacking:

- Optimal learning rate schedules for accelerating grokking
- Adaptive weight decay strategies
- Curriculum learning for inducing grokking

7.2.2 Novel Optimizers

Work [20] mentions that recent optimizers like Muon may be effective for grokking. Future research:

- Comparative analysis of optimizers (Adam, AdamW, SGD, Muon, Lion, etc.)
- Development of optimizers specifically designed for grokking
- Adaptive switching between optimizers in different phases

7.3 Cross-Domain Applications

7.3.1 Scientific Computing

Potential applications:

- Training neural networks to solve PDEs with grokking to correct physical solutions
- Molecular dynamics: grokking to conservation laws
- Climate models: delayed generalization to long-term trends

7.3.2 Medicine and Biology

- Training on small medical datasets with grokking
- Discovering biological mechanisms through mechanistic interpretation of grokked models
- Transfer learning from grokked models for rare diseases

7.4 AI Safety and Alignment

7.4.1 Interpretability

Grokking provides unique opportunity for mechanistic interpretability:

- Can compare model before and after grokking
- Identify which components change
- Extract interpretable algorithms

Open questions:

1. Can we induce grokking toward more interpretable solutions?
2. Are grokked models inherently more interpretable?
3. How to use grokking for automated circuit discovery in large models?

7.4.2 Unlearning and Privacy

As shown in Section 5.4, grokked models are better unlearners. Future directions:

- Methods for inducing grokking specifically to improve unlearning
- Certified data deletion through grokking
- Fairness: ensuring uniform unlearning for minorities

7.5 Fundamental Questions**7.5.1 Universality of Grokking**

- Is grokking a universal phenomenon in all learning systems?
- Does grokking occur in biological neural networks?
- Can grokking be observed in human learning?

7.5.2 Connection to Cognitive Science

Intriguing parallels:

- “Aha!” moments in human problem-solving
- Incubation effects in creativity
- Memory consolidation during sleep

Question: Is grokking a computational model of insight?

8 Conclusion

The grokking phenomenon, first described just four years ago, has led to an explosion of research activity, spawning over 80 publications and deeply enriching our understanding of neural network training dynamics.

8.1 Key Takeaways

1. Theoretical understanding:

- Grokking can be understood as phase transition, norm minimization on zero-loss manifold, or transition from lazy to rich learning
- Implicit regularization and implicit bias play critical roles
- Provable scaling laws and critical data thresholds exist

2. Mechanistic understanding:

- Models learn structured algorithms (e.g., DFT for modular arithmetic)
- Three phases: memorization, circuit formation, cleanup
- Flatness necessary for generalization; neural collapse role remains debated

3. Practical methods:

- 50+ times grokking acceleration possible (Grokfast, GrokAlign, embedding transfer)
- Grokking applicable to real-world tasks (LLMs, RL, knowledge graphs)
- Grokked models possess desirable properties (better unlearning, interpretability)

4. Connections to other phenomena:

- Unification with double descent, benign overfitting
- Connection to lottery ticket hypothesis, emergent abilities
- Potential parallels with cognitive processes

8.2 Broader Implications

Grokking challenges traditional views on:

- **When to stop training:** Early stopping on test error may miss subsequent grokking
- **Regularization:** Weight decay acts not only through explicit penalization but also through shaping training dynamics
- **Overfitting:** Not always harmful; can be intermediate phase to better generalization
- **Scaling:** More training (beyond overfitting) can qualitatively change learned solutions

8.3 Final Reflections

Grokking is a reminder that deep learning remains full of surprises. Despite decades of research, fundamental dynamics of neural networks continue to reveal new phenomena.

Intensive study of grokking over the past four years demonstrates the power of interdisciplinary approach: theoretical analysis, mechanistic interpretability, empirical investigations, and practical applications mutually enrich each other, creating an increasingly complete picture.

As models scale to trillions of parameters and train on increasingly complex tasks, phenomena like grokking may become not the exception but the rule. Understanding these phenomena is not just academic interest, but necessity for building reliable, interpretable, and safe AI systems.

Grokking teaches us patience: sometimes the deepest understanding comes not immediately, but after sustained effort. This is true both for neural networks and for the researchers studying them.

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